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RESEARCH ARTICLE

CBCTL-IDS: A Transfer Learning-Based Intrusion Detection System Optimized With the Black Kite Algorithm for IoT-Enabled Smart Agriculture

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ABSTRACT Smart Agriculture(SA) stands as a critical application frontier for the evolution of the IoT, catalyzing the transformation of traditional agricultural practices into modern, data-driven, intelligent, and automated systems. This transition not only elevates the levels of automation and intelligence in agricultural production but also introduces a spectrum of cybersecurity challenges. In particular, the widespread deployment of interconnected devices within these systems poses significant risks to the security and integrity of data. As a pivotal defense mechanism against malicious activities and anomalous traffic, Network Intrusion Detection Systems (NIDS) play an indispensable role in ensuring the security of cyberspace. Nevertheless, existing methods for detecting anomalous traffic in IoT networks still exhibit limitations in accuracy, thereby hindering the efficiency of identifying potential threats. To enhance the capability of threat detection, this paper proposes a novel network intrusion detection method, CBCTL-IDS, based on transfer learning. This method integrates three core components: Convolutional Neural Networks (CNN), the Black Kite Algorithm (BKA), and a Confidence Averaging mechanism. By incorporating pre-trained models such as MobileNet, EfficientNet, Xception, VGG19, and Inception into the domain of IoT intrusion detection, CBCTL-IDS leverages transfer learning to achieve efficient feature extraction and classification, addressing the limitations of traditional methods in feature representation. Furthermore, the BKA is utilized to adaptively optimize the hyperparameters of multiple DL models, significantly enhancing their classification performance. Through ensemble learning, the top three performing models are selected, and the Confidence Averaging mechanism is applied to further improve the stability and reliability of detection results. Experimental results demonstrate that the CBCTL-IDS method achieves detection accuracy rates exceeding 99% on three IoT intrusion detection datasets: ToN-IoT, Edge-IIoTset, and WSN-DS significantly outperforming existing mainstream methods. This approach provides robust technical support for the security protection of IoT systems.

INDEX TERMS Intrusion detection system, smart agriculture, Internet of Things, transfer learning.

I. INTRODUCTION

According to relevant research reports, approximately 500 million people worldwide are affected by malnutrition, while 821 million face the challenge of hunger [1]. Consequently, identifying effective methods to enhance the crop yield per unit area has become increasingly urgent. Smart Agriculture(SA), by integrating information technologies, data analytics, and intelligent decision-making control mechanisms, not only improves agricultural production efficiency and product quality but also provides a viable solution to address food shortages [2]. However, the widespread use of interconnected devices in SA systems has also made them a prime target for cybersecurity threats, creating opportunities

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for potential attackers. In recent years, agricultural systems have frequently been subjected to cyberattacks. For instance, in 2017, Maersk, a global leader in shipping and logistics, suffered a severe disruption to its operations due to the NotPetya malware attack [3]. By 2021, this trend showed no signs of abating, with six major agricultural organizations falling victim to ransomware attacks [4]. These attacks not only disrupted the supply chains of critical agricultural inputs such as seeds and fertilizers but also had a significant negative impact on normal crop harvesting. These cases underscore the urgency and importance of strengthening the security defenses of SA systems. The Intrusion Detection System (IDS) is one of the pivotal technologies in the field of cybersecurity. By monitoring abnormal patterns in network traffic in real time, it can effectively identify potential network threats or illicit activities, serving as a critical mechanism for defending against cyberattacks [5]. IDS are primarily categorized into two types: (1) Network Intrusion Detection Systems (NIDS) and (2) Host-based Intrusion Detection Systems (HIDS) [6]. NIDS are typically deployed at critical network nodes to ensure coverage of more vulnerable traffic points, while HIDS operate on connected devices to safeguard the devices themselves. Given the vast number of IoT devices, developing independent security solutions for each device is highly challenging. Therefore, protecting network transmissions within IoT systems has emerged as a feasible strategy [7]. This paper focuses on NIDS.

Currently, in response to network threats targeting IoT-enabled smart systems, researchers have developed IDS in domains such as SA [8], smart cities [9], and smart healthcare [10] to ensure the security of these systems. However, existing IoT IDS based on Machine Learning (ML) and Deep Learning (DL) still face several challenges. First, the training data is often outdated and fails to reflect the characteristics of the current IoT network environment. Second, existing IDS technologies exhibit limited overall performance when addressing multi-class classification problems involving imbalanced datasets. To address the aforementioned challenges, this paper proposes the CBCTL-IDS method. First, the Extremely Randomized Trees(ET) algorithm is employed to extract key features, and the dataset is transformed into image sets for model input. Subsequently, the Black Kite Algorithm (BKA) is utilized to optimize the hyperparameters of five Convolutional Neural Networks (CNN) models: MobileNet, EfficientNet, Xception, VGG19, and Inception. Finally, the top three CNN models with the best performance are selected, and ensemble learning is achieved through a confidence averaging strategy. To validate the effectiveness of the proposed method, performance analysis and experimental validation are conducted on the IoT datasets ToN-IoT and Edge-IIoTset, as well as the sensor network dataset WSN-DS. The main contributions of this paper are summarized as follows:

- Tabular data is transformed into image blocks and fed into CNN models utilizing deep transfer learning. By employing a confidence averaging strategy to integrate the top three best-performing models, the classification performance of the model is significantly enhanced.
- Common threats in smart agriculture are systematically analyzed, and the ToN-IoT, Edge-IIoTset, and WSN-DS datasets are used to simulate the network traffic characteristics of Smart Agricultural(SA) environments. Among these, the ToN-IoT dataset adopts a balanced strategy, while the WSN-DS and Edge-IIoTset datasets retain their original data distributions to evaluate the model's performance on both balanced and imbalanced datasets. Experimental results demonstrate that the model achieves an accuracy of 99% across all three datasets.
- The BKA is introduced to optimize the hyperparameters of CNN models, leading to a notable improvement in classification performance.

The structure of this paper is organized as follows: Section II explores the network threats faced by SA and reviews related research on IDS. Section III provides a detailed description of the proposed IDS method. Section IV presents the performance analysis and evaluation of the proposed scheme. Section V concludes the paper, discusses its limitations, and outlines future research directions.

II. RELATED WORK

A. NETWORK THREATS IN SMART AGRICULTURE

SA represents an extension of IoT technologies, Software-Defined Networking (SDN), 5G communications, and cloud computing into the agricultural domain, enabling more precise and intelligent management of agricultural production [6]. However, the open communication environment of SA systems also renders them potential targets for cyberattacks, exposing them to a wide array of network threats. Table 1 summarizes the common network threats in SA. Attackers employ various methods to gain unauthorized access to network systems, including but not limited to Denial of Service (DoS), Distributed Denial of Service (DDoS), Man-in-the-Middle (MITM) attacks, ransomware attacks, and injection attacks. These attacks not only pose significant risks to the confidentiality, integrity, and availability of data but may also result in substantial losses to agricultural production.

Hardware devices in SA, such as drones, agricultural robots, and sensors, can lead to significant economic losses or even criminal activities if compromised by attackers. For instance, damage to cultivated land or uncontrolled drones colliding with buildings are potential consequences. Common attack vectors include GPS spoofing, side-channel attacks, and blackhole attacks [2], [8], [11], [12], [13]. In the realm of smart agriculture, networks are frequently subjected to threats posed by DDoS and DoS attacks [14], [15]. These attacks are typically orchestrated by malicious entities leveraging botnets to inundate the connected devices within agricultural systems with an overwhelming volume

TABLE 1. Security threats and vulnerabilities in smart agriculture systems.

Attack Target	Attack Category	Violated Security Aspect	References
Smart agriculture hardware	Side Channel Attack, Physical Attacks, Blackhole	Confidentiality, Availability	[2], [8], [11], [12], [13]
Smart agriculture network	Distributed Denial-of-Service, Denial of Service, Man-In-The-Middle Attacks, Botnets	Confidentiality, Integrity, Availability	[2], [14], [15], [16], [17], [18]
Smart agriculture data and information	Cloud Computing Attacks, Data Leakage, False Data Injection, SQL Injection Attack, Ransomware, Scanning, Theft	Confidentiality, Integrity, Availability	[2], [19], [20], [21]

of requests, thereby rendering the devices inoperable. Furthermore, attackers often exploit protocol vulnerabilities to execute ARP spoofing and MITM attacks [2], [16], [18], which further compromise the integrity and security of network communications. In the context of smart agriculture, data and information are typically stored within servers and computational platforms. Competitors or adversarial nations may seek to analyze sensitive information such as crop yields, pricing, and production volumes to gain a strategic advantage in commercial or international competition. Consequently, such information holds significant appeal for malicious actors. Common attack methodologies include scanning attacks and data leakage attacks [2], [20]. Additionally, data injection attacks and targeted assaults on cloud computing platforms pose substantial threats to agricultural data servers [19], [21].

B. IDS

The IDS stands as a critical security technology, capable of identifying and isolating malicious intrusions while efficiently monitoring illicit activities and anomalous traffic within IoT systems [5]. In recent years, with the escalating severity of security threats targeting IoT systems, research on intrusion detection systems for IoT has emerged as a prominent area of focus, garnering widespread attention. Researchers have proposed a variety of effective solutions to address these challenges.

In the domain of IoT security, Li et al. [22] proposed an intrusion detection methodology based on multi-CNN fusion, which was subsequently evaluated on the NSL-KDD dataset. Compared to Recurrent Neural Networks (RNN), Deep Convolutional Neural Networks (DCNN), and Support Vector Machines (SVM), the multi-CNN fusion model demonstrated superior performance in terms of detection accuracy and computational complexity. In a related study, Kasongo [23] introduced an advanced IDS leveraging genetic algorithms and tree-based algorithms, designed to enhance the security of Industrial IoT (IIoT). Evaluated on the UNSW-NB15 dataset, this model achieved a test accuracy of 87.61% for binary classification tasks. Furthermore, Sharma et al. [24] proposed an Anomaly-based IDS utilizing Deep Neural Networks (DNN) to classify IoT network attacks, attaining an accuracy of 84% on the UNSW-NB15 dataset. Additionally, Elnakib et al. [25] developed an Enhanced Intrusion Detection Model (EIDM) based on DL, specifically tailored to detect and classify 15 types of traffic behaviors, including 14 attack types, within the CIC-IDS-2017 dataset. The EIDM achieved an impressive accuracy of 95% for the 15-class attack classification task, outperforming existing IDS models. Aslam et al. [26] propose a hybrid network intrusion detection system (Hybrid NIDS) that integrates rule-based matching (SNORT) and machine learning classification to mitigate the high false positive and false negative rates inherent in traditional rule-based systems such as SNORT. Experimental evaluation on the KDD99 dataset demonstrates that the proposed system achieves a detection rate of 97.86% while reducing the false alarm rate to 2.11%. However, the aforementioned IDS solutions were evaluated solely on a single dataset, and the KDD99, NSL-KDD, UNSW-NB15, and CIC-IDS-2017 datasets fail to adequately capture the characteristics of contemporary IoT network attacks. Khan et al. [27]introduced an IDS based on Long Short-Term Memory (LSTM) autoencoders, designed to enhance the security of modern industrial control systems. The model's performance was evaluated on two publicly available datasets, achieving accuracies of 97.95% and 97.62% on the gas pipeline dataset and the NSW-NB15 dataset, respectively. In a related study, Lilhore et al. [28] proposed a Hybrid Intrusion Detection Model (HIDM) for safeguarding Industry 4.0 networks. This model integrates an optimized CNN-LSTM architecture with transfer learning and employs the Grey Wolf Optimization (GWO) algorithm to fine-tune CNN parameters, thereby improving prediction accuracy. Evaluated on the ToN-IoT and UNSW-NB15 datasets, the model achieved precision rates of 92.7% and 94.25%, respectively. Nevertheless, there remains significant room for further enhancement in the performance of existing IDS solutions.

In recent years, the technique of transforming one-dimensional network traffic data into images has gradually gained prominence and has been extensively applied to CNN models based on transfer learning, demonstrating exceptional performance. El-Ghamry et al. [8]proposed a DL model based on transfer learning, utilizing the VGG16 architecture to detect anomalous traffic in smart agriculture. This model achieved a detection accuracy of 99.99% on the NSL-KDD dataset, providing robust support for the security of SA systems. However, this method was only tested on a single dataset, lacking validation for broader applicability. Toldinas et al. [29] introduced a DNN+ResNet50 approach, which converts network security datasets into four-channel images and leverages ResNet50 pre-trained weights for deep transfer learning. This method achieved detection accuracies of 99.80% and 99.70% on the UNSW-NB15 and BOUN DDoS datasets, respectively. The ELETL-IDS [30] framework, by integrating five pre-trained CNN models including VGG16, Inception, and EfficientNet, achieved a 100% detection accuracy on the CIC-IDS2017 and CSE-

CICIDS2018 datasets. Similarly, the TL-CNN-IDS [31] scheme proposed transforms non-image formatted network intrusion data into images and employs transfer learning with pre-trained weights from VGG16, Inception, and Xception, achieving commendable performance on both the CIC-IDS-2017 and NSL-KDD datasets. Nevertheless, despite the outstanding performance of these deep transfer learning-based intrusion detection schemes, the experimental datasets used are relatively outdated, making it difficult to reflect the complexities and dynamics of contemporary network environments.

Smart agriculture, as an application of IoT technology in the agricultural sector, places paramount importance on network security. The development of high-precision IDS to defend against various network threats has become an urgent issue to address [32]. However, existing IDS solutions for IoT systems still face numerous challenges, such as suboptimal model accuracy, poor data quality, and a lack of broad applicability. To address these issues, this paper proposes CBCTL-IDS, which selects IoT security datasets including ToN-IoT, wireless sensor network dataset WSN-DS, and IoT intrusion detection dataset Edge-IIoTset. The BKA algorithm is employed to optimize the hyperparameters of five CNN models MobileNet, EfficientNet, Xception, VGG19, and Inception and the top three models with the best performance are selected to construct the final classifier. Experiments are conducted and validated on both balanced and original imbalanced datasets, with results demonstrating that the proposed method exhibits robust detection performance.

III. CBCTL-IDS

CBCTL-IDS aims to develop a NIDS capable of effectively identifying security threats within SA networks, thereby ensuring their security. The proposed IDS architecture is illustrated in Figure 1. The proposed IDS is deployed at the gateway of the SA system, primarily based on the following considerations: First, gateway devices possess substantial computational capabilities and network efficiency when executing edge computing tasks, enabling them to support large-scale model inference and real-time data processing [32]. Second, as a critical traffic node within the SA system, the gateway serves as the primary interface through which the majority of network traffic interacts with external environments [33]. This deployment strategy facilitates the efficient identification of malicious traffic within the SA network by the IDS. The proposed CBCTL-IDS in this paper is an IDS that integrates BKA-optimized CNN and transfer learning, designed to detect network threats in smart agriculture. Figure 2 provides an overview of the proposed IDS framework. Network datasets from IoT sensor networks and gateways undergo a data preprocessing phase. Duplicate values and missing values (NaN) are removed, followed by the extraction of features highly correlated with attack categories using the ET algorithm. Subsequently, the data is transformed into images using the quantile transformation method. In the next stage, the generated image sets are trained

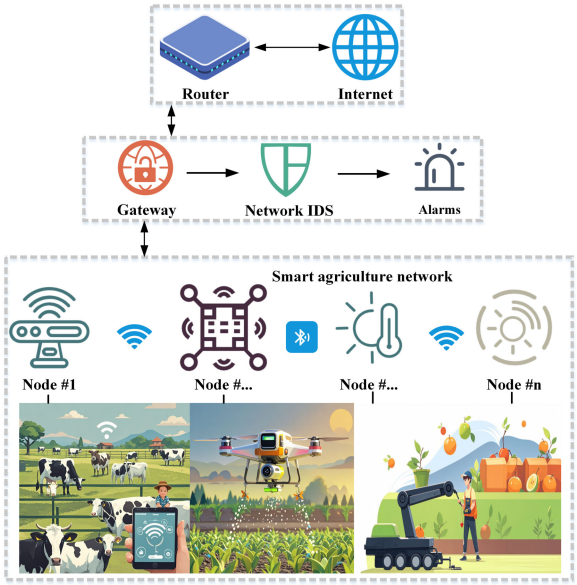

FIGURE 1. The proposed IDS framework protects Smart agriculture.

using five CNN models MobileNet, EfficientNet, Xception, VGG19, and Inception to construct base learners. These CNN models are optimized by the BKA algorithm, which automatically adjusts hyperparameters. The top three best-performing CNN models are then selected as the base models to build an ensemble learning model. Finally, confidence averaging is employed to construct the final detection output of the ensemble model.

A. DATASETS AND DATA PROCESSING MODULE

To develop an IDS capable of detecting network threats in smart agriculture, three datasets were utilized in this work: the WSN-DS dataset, generated from a heterogeneous sensor network [19], and the ToN-IoT datasets [18] and Edge-IIoTset [21], All three datasets are derived from real IoT environments and encompassing the latest attack patterns in IoT settings. These three datasets ToN-IoT, WSN-DS, and Edge-IIoTset cover a wide range of components, including sensor nodes, edge devices, and cloud servers, making them highly representative. They effectively capture the characteristics of SA environments, thereby providing reliable data support for evaluating the performance of the IDS. The ToN-IoT dataset represents a next-generation dataset designed for Industrial 4.0 and IoT scenarios. The testbed was deployed using multiple virtual machines and physical hosts running Windows, Linux, and Kali operating systems to manage the interconnections among the IoT, cloud, and edge/fog system layers [18]. In this study, the ToN-IoT_Network subset was utilized, which comprises 44 flow features/attributes and includes 9 distinct types of attack traffic (e.g., Injection, MITM, BackDoor, DDoS, DoS, Ransomware, Scanning, XSS, and Password) as well as normal traffic, totaling 10 traffic categories. The dataset consists of 50,000 normal instances and 161,043 attack instances, with their distribution

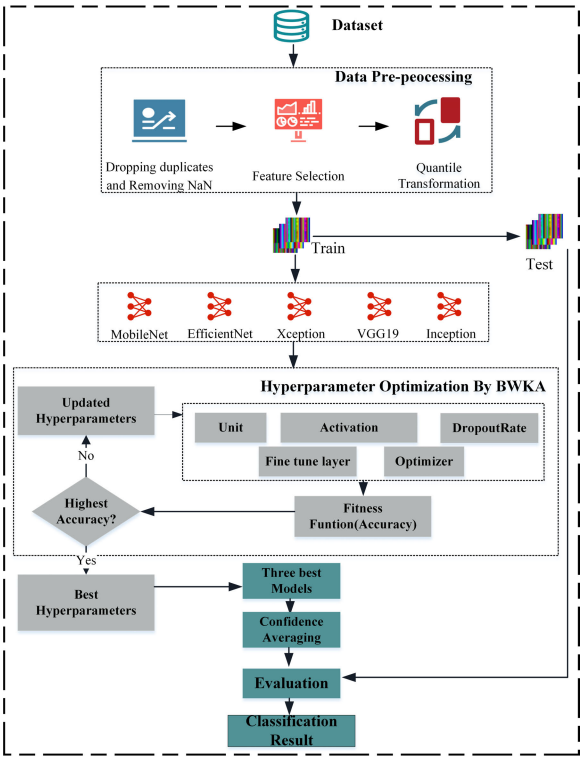

FIGURE 2. The architecture of the proposed IDS.

illustrated in Figure3. Features of the Object type were encoded, and 27 non-friendly character-based features (such as 'dns_AA' and 'dns_RA') were numerically encoded [34].

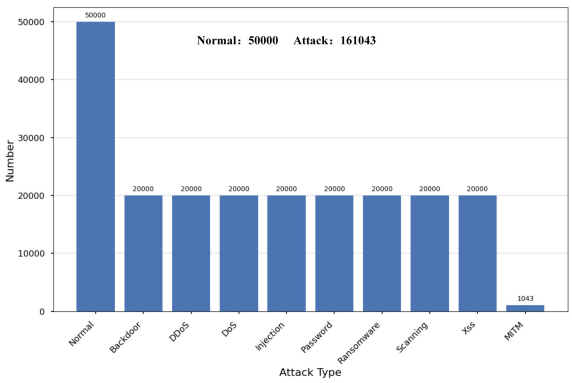

FIGURE 3. Distribution of data categories in the ToN-IoT dataset.

The WSN-DS dataset was developed by Almomani et al. with the aim of enhancing the detection and classification capabilities for various attacks in IoT environments. The dataset was generated from a heterogeneous sensor network [19]. WSN-DS encompasses four types of attacks (including Grayhole, Blackhole, Flooding, and TDMA) as well as normal traffic, comprising 23 attributes to identify the state of each node in the network. The dataset includes 340,066 normal instances and 34,695 attack instances, with their distribution illustrated in Figure4. Unlike other datasets, WSN-DS does not contain features of the Object type, thus eliminating the need for numerical encoding.

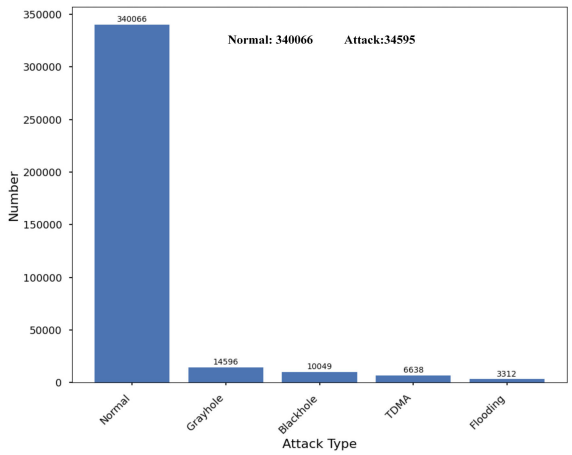

FIGURE 4. Distribution of data categories in the WSN-DS dataset.

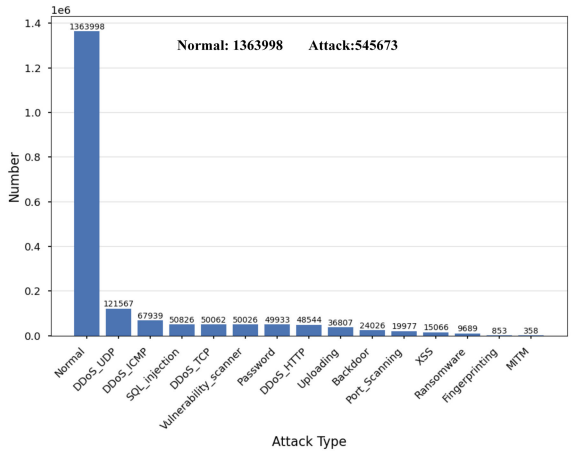

FIGURE 5. Distribution of data categories in the Edge_IIoT dataset.

The Edge-IIoTset dataset was generated using a dedicated IoT/IIoT testbed equipped with a wide range of representative devices, sensors, protocols, and cloud/edge configurations. This dataset encompasses 14 attack categories related to IIoT connectivity protocols [21]. During the data preprocessing phase, 815 duplicate rows were removed, and 16 unnecessary features, such as 'frame.time', 'ip.dst_host', 'ip.src_host', and 'arp.src.proto_ipv4', were eliminated [35]. Subsequently, character-based features were numerically encoded. The preprocessed Edge-IIoTset dataset comprises 1,909,671 rows and 96 columns, with its distribution illustrated in Figure 5. To evaluate the robustness and applicability of the model, this study conducts tests on both balanced and original imbalanced datasets. For the ToN-IoT dataset, the SMOTE technique is employed for oversampling, expanding the samples of the MitM category to 20,000 while retaining the first 20,000 samples of the "Normal" category. The

WSN-DS and Edge-IIoTset datasets, however, maintain their original imbalanced class distributions. All datasets are partitioned into 70% for training, 10% for validation, and 20% for testing to ensure a comprehensive evaluation of the model's performance.

B. FEATURE ENGINEERING MODULE

Datasets often contain a large number of features. Irrelevant features increase the cost of model training and computational resource consumption. Therefore, feature selection is used to retain key features and reduce data redundancy. The ET algorithm calculates the importance of each feature and identifies those that contribute most to model prediction. These importance scores assist in feature selection, thereby reducing dimensionality and enhancing model performance [36]. The importance of a feature is determined by calculating the average reduction in Gini impurity across the entire random forest.

$$G = \sum_{i=1}^{c} P_i(1 - P_i) \tag{1}$$

Among them, C is the number of categories, and P_i is $i$the probability of belonging to the i th category.

$$\triangle G = G_{before} - G_{after} \tag{2}$$

For each tree, the Gini impurity reduction for each feature at all nodes is calculated, as described in Equation (1). The reductions are then summed to obtain the total reduction for each feature in the tree. Finally, the feature importance for each tree is summed and divided by the total Gini impurity reduction to compute the feature importance score. The calculation principle is further detailed in Equation (2). Table 2 presents the key features retained after applying the ET algorithm to the ToN-IoT and WSN-DS datasets. For the ToN-IoT dataset, 20 features highly correlated with the labels were retained, characterized by a Gini impurity reduction greater than 0.0062. In the case of the WSN-DS dataset, 9 features strongly associated with the labels were preserved, indicating a Gini impurity reduction exceeding 0.0474. As for the Edge-IIoTset dataset, 24 network features highly correlated with the labels were retained, with their Gini impurity reduction exceeding 0.015712. In the ToN-IoT dataset, the ET algorithm was employed to identify 20 features exhibiting a strong correlation with the target labels. Originally, this dataset comprised 44 flow-related attributes extracted from network packet capture (pcap) files via the Zeek tool [18]. Table 2 delineates the 20 network features retained after feature selection. For example, the "dns_AA" attribute, a Boolean variable, signifies an authoritative answer in DNS resolution, where "T" indicates that the server possesses authoritative control over the queried domain. Similarly, "dns_RA" represents recursion availability, with "T" denoting that the server supports recursive queries. By leveraging the ET algorithm for feature engineering, the dimensionality of the ToN-IoT, WSN-DS, and Edge-IIoTset datasets was reduced by more than half, thereby improving computational efficiency while preserving the most salient network features critical for intrusion detection.

C. DATA CONVERSION

Given the superior performance of CNN models on image-based datasets and the inherently tabular nature of network traffic data, it is essential to transform raw network data into an image representation [37]. To achieve this, pixel values within the generated image blocks are constrained to the range of 0 to 255. The Quantile Transformation (QT) method [38] is utilized to rescale network traffic data onto this predefined scale, ensuring a more suitable input representation for CNN-based processing. The mathematical formulation of QT is delineated in Equation (3)

$$X = (X - Min(X))/(Max(X) - Min(X)) \times 255 \tag{3}$$

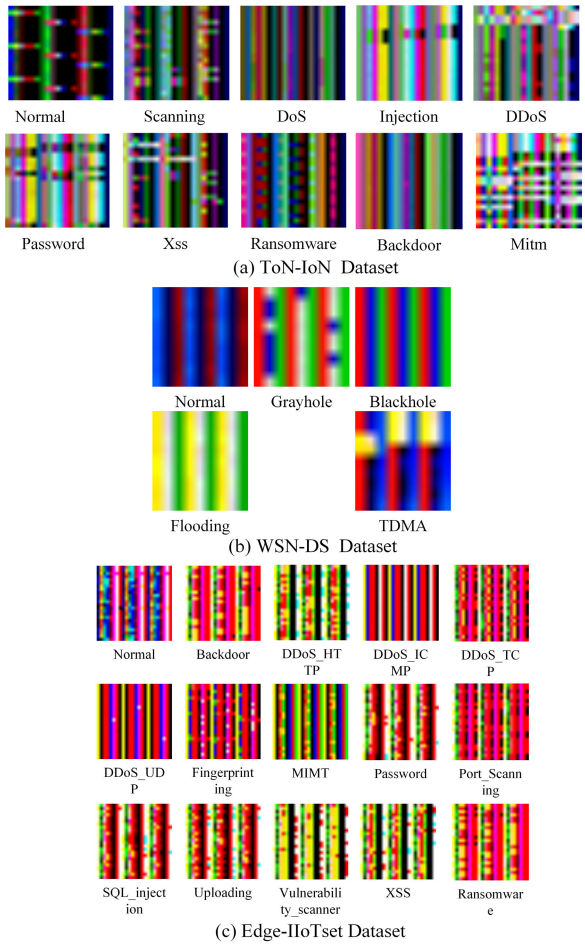

FIGURE 6. Distribution of tabular data transformation into image sets.

Following normalization, data samples are segmented into structured blocks. In the ToN-IoT dataset, after feature engineering, each set of 60 consecutive homogeneous samples(each comprising 20 attributes)is reshaped into a 20×20×3 image block, encapsulating a total of 1,200 feature values. Similarly, for the feature-selected WSN-DS dataset, each group of 27 contiguous samples, with

TABLE 2. Feature selection by applying ET.

Dataset	Features
ToN-IoT	dns_query, dst_bytes, src_bytes, dns_AA, dns_RA, dns_rejected, dst_ip_bytes, dst_pkts, src_pkts, src_ip_bytes, service, dst_ip, duration, src_ip, proto, src_port, dst_port, conn_state, ts
WSN-DS	Is_CH, Expaned, Energy, JOIN_S, Data_Sent_To_BS, dist_CH_To_BS, ADV_R, DATA_S, Time, SCH_S
Edge-IIoTset	mqtt.conack.flags-0.0, mqtt.topic-0.0, tcp.ack, mqtt.topic-0, tcp.checksum, icmp.checksum, mqtt.protoname-0.0, tcp.seq, tcp.ack_raw, dns.qry.name.len-0.0, tcp.len, dns.qry.name.len-0, tcp.flags, http.referer-0, udp.stream, http.request.version-0.0, icmp.seq_le, http.request.method-0.0, http.referer-0.0, mqtt.protoname-0, mqtt.conack.flags-0, tcp.flags.ack, http.request.version-0, http.request.method-0

9 attributes per sample, is transformed into a 9×9×3 image block, encapsulating 243 feature values. Meanwhile, in the feature-engineered Edge-IIoTset dataset, every set of 72 sequential homogeneous samples, each containing 24 attributes, is converted into a 24×24×3 image block, encompassing 1,728 feature values. The resulting image blocks are three-channel square color images (red, green, and blue). Taking the ToN-IoT dataset as an illustrative case, the first 400 samples within each block are allocated to the first channel of the image matrix, the subsequent 400 samples to the second channel, and the final 400 samples to the third channel, thereby ensuring all samples are systematically mapped onto the RGB channels. The transformation from structured tabular data to image matrices is efficiently executed using the OpenCV library. The three datasets WSN-DS, ToN-IoT, and Edge-IIoTset were transformed into corresponding image datasets, which serve as inputs for the proposed model. The representative image blocks for each class in the ToN-IoT dataset are illustrated in Fig 6(a), while the WSN-DS dataset's representative images, following transformation, are presented in Fig 6(b). For the Edge-IIoTset dataset, the converted image representations distinctly capture the inherent characteristics of different traffic types, as depicted in Fig 6(c). For example, in the WSN-DS dataset, image blocks associated with "Normal" traffic exhibit a structured striped pattern, whereas those corresponding to "Grayhole" attacks display a combination of dispersed points and striped formations, reflecting the underlying network behavior differences.

D. CNN AND DEEP TRANSFER LEARNING MODULE

CNNs have been widely applied in image processing tasks. Once network traffic is transformed into images, it can be directly fed into a CNN architecture for processing. A typical CNN architecture consists of three main components [39]: convolutional layers, pooling layers, and fully connected layers. Convolutional layers extract local features from the input data using convolutional kernels, while pooling layers reduce the dimensionality of the features while retaining critical information. The fully connected layers integrate the high-level features extracted by the preceding layers and produce the final output through nonlinear transformations.

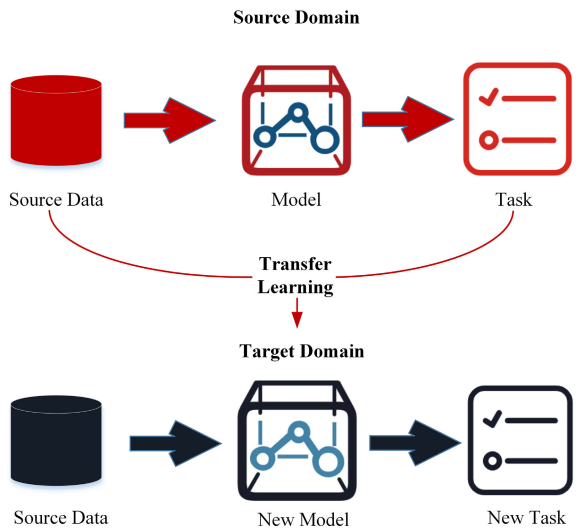

FIGURE 7. Schematic diagram of transfer learning principles.

The proposed IDS is based on transfer learning techniques, which involve pre-training a model on one dataset and then applying it to another dataset for a different task [40]. The fundamental principle is illustrated in Figure 7: by pre-training on source domain data, the model can learn general feature representations. Subsequently, the pre-trained model is transferred to the target domain, and its parameters are fine-tuned based on the specific characteristics of the target domain to adapt to the new task requirements. During the fine-tuning process, most of the lower-level parameters are frozen, while only a portion of the top layers are unfrozen to learn the features of the new dataset. This fine-tuning approach, by appropriately adjusting the model parameters, enables the model to better learn the features of the new dataset, thereby enhancing the performance of the deep transfer learning (DTL) model [41]. Deep transfer learning (DTL) has been successfully applied to intrusion detection tasks. Since pre-trained models have already learned general feature representations, fine-tuning on target tasks can significantly accelerate the training and convergence process [42], [43].

In this study, five models: MobileNet, EfficientNet, VGG19, Inception, and Xception were optimized using the BKA algorithm for hyperparameter tuning, and the top three models with the best performance were selected for ensemble learning using confidence averaging. These models have been pre-trained on the ImageNet dataset and have demonstrated exceptional performance in the field of image classification. MobileNet, proposed by Google in 2017, is a lightweight model that employs depthwise separable convolution to significantly reduce model size and computational complexity while maintaining high accuracy [44]. EfficientNet utilizes the AutoML-MNAS framework for neural architecture search and employs scaling techniques to adjust the network's width, depth, and resolution [45]. VGG19, developed by the Visual Geometry Group at the University of Oxford, consists of 16 convolutional layers and is renowned for its simple yet effective architectural design, achieving remarkable results in image classification tasks [46]. The Inception architecture, introduced by Google in 2014, is based on the core idea of capturing multi-scale information in images by using parallel convolutional kernels of varying sizes, thereby enhancing the model's expressiveness and efficiency [47]. Xception, proposed by François Chollet in 2017, is an efficient CNN architecture that builds upon the Inception model and further improves model efficiency and performance by incorporating depthwise separable convolution [48]. The proposed deep transfer learning-based IDS in this study was extensively evaluated on the IoT datasets ToN-IoT and Edge-IIoTset, as well as the wireless sensor network dataset WSN-DS.

E. BLACK-WINGED KITE ALGORITHM

Hyperparameter selection is critical to the performance and efficiency of the model, and manual tuning is both time-consuming and inefficient. In this study, the BKA was employed as a hyperparameter optimization algorithm to enhance model accuracy. BKA performs hyperparameter optimization by simulating the three primary behavioral modes of black kites in grasslands: population initialization, attack behavior, and migration behavior [49]. The BKA exhibits robust evolutionary capabilities, rapid search speed, and strong optimization performance. Its unique bio-inspired characteristics endow it with powerful dynamic search capabilities [50].

1) POPULATION INITIALIZATION

First, evenly distribute the position of each black-winged kite. The initialization principle is as Equation (4):

$$X_i = BK_{lb} + rand(BK_{ub} - BK_{lb}) \tag{4}$$

For an integer i ranging from 1 to pop, let BK_{lb} and BK_{ub} denote the lower and upper bounds for the ith Black-winged kite in the jth dimension, respectively, with *rand* being a randomly selected value within the interval [0, 1]. During the initialization phase, the BKA algorithm selects the individual with the smallest fitness value, designated as XL, to assume the role of group leader. The determination of XL follows the calculation principles outlined in equations (5) and (6):

$$f_{best} = min(f(X_i)) \tag{5}$$

$$XL = X(find(f_{best} == f(X_i)) \tag{6}$$

2) ATTACK BEHAVIOR

Black-winged kites, being predators of grassland ecosystems, exhibit two distinct behaviors during hunting: they either hover in the air to scout for potential prey, representing a state of global exploration, or swiftly dive towards their target once prey is located, which models an exploitation search. The mathematical formulation that captures the principle of the attack behavior of black-winged kites is presented in Equation (7):

$$y_{t+1}^{i,j} = \begin{Bmatrix} y_t^{i,j} + n(1 + sin(r)) \times y_t^{i,j}, p < r \\ y_t^{i,j} + n \times (2r - 1) \times y_t^{i,j}, else \end{Bmatrix} \tag{7}$$

$$n = 0.05 \times e^{-2\times(t/T)^2} \tag{8}$$

where $y_t^{i,j}$ and $y_{i,j}^{t+1}$ denote the position of the ith Black-winged kite in the jth dimension at the tth iteration and the $(t+1)^{th}$ iteration, respectively. r is a randomly generated number in the range [0,1], while p is set to a constant value of 0.9, T represents the total number of iterations, whereas t denotes the current iteration number.

3) MIGRATION BEHAVIOR

Black-winged kites, as migratory birds, have their movement patterns significantly influenced by environmental conditions. During migration, a leader typically guides the group. The migration behavior can be divided into two scenarios based on fitness comparisons: If the fitness of the current population is inferior to that of a randomly selected population, the existing leader steps down and joins the migrating flock, indicating its inability to effectively lead. Conversely, if the current population's fitness surpasses that of the random population, the leader continues to guide the migration process. The mathematical representation of this principle is provided in Equation (9).

$$y_{t+1}^{i,j} \begin{cases} y_t^{i,j} + C(0,1) \times (y_t^{i,j} - L_t^j), F_i < F_{ri} \\ y_t^{i,j} + C(0,1) \times (L_t^j - m \times y_t^{i,j}), else \end{cases} \tag{9}$$

$$m = 2 \times sin(r + \pi/2) \tag{10}$$

L_t^j denotes the leading scorer of the Black-winged kites in the jth dimension up to the tth iteration. $y_t^{i,j}$ and $y_{t+1}^{i,j}$ denote the position of the ith Black-winged kite in the jth dimension at the t and $(t+1)^{th}$ iteration steps, respectively. F_i denotes the position in the jth dimension achieved by any Black-winged kite during the t iteration. F_{ir} denotes the fitness value of a random position in the jth dimension obtained by any Black-winged kite during the ith iteration. $C(0,1)$ denotes the Cauchy mutation [51].

The BKA algorithm is employed to optimize the hyperparameters of the CNN model, including Units, Activation, Dropout rate, Fine-tuning layers, and Optimizer. The range for Units is [128, 256], for Activation is ['relu', 'selu', 'elu'], for Dropout rate is [0.1, 0.6], and for Optimizer is ['adam', 'nadam', 'adamax', 'adagrad']. The values for Frozen layers are specified in Table 3 for each model. Taking VGG19 as an example, after BKA optimization, the Units are set to 256, Activation to 'relu', Dropout rate to 0.6, Fine-tuning layers to 19, and Optimizer to 'adam'.

TABLE 3. Hyperparameter optimization for model fine-tuning via the BKA.

Model	Frozen layers	Unis	Activation	Dropout rate	Optimizer
Inception	[30, 50],36	256	elu	0.5	adamax
Xception	[100, 150],124	256	relu	0.3	nadam
VGG19	[10, 19],19	128	selu	0.6	nadam
MobileNet	[10, 30],12	256	selu	0.3	adamax
EfficientNet	[270, 350],297	128	elu	0.2	adamax

F. CONFIDENCE AVERAGING ENSEMBLE MODULE

The confidence averaging strategy is a commonly employed technique in ensemble learning. In DL models, the softmax layer generates a posterior probability distribution that represents the classification confidence for each class. The confidence averaging method computes the mean classification probability of each class across all base learners and selects the class with the highest average confidence score as the final predicted label [52]. By averaging the posterior probabilities assigned by individual base learners to each class, the confidence averaging approach ensures a more robust classification outcome. The final class label is determined based on the highest mean confidence score. Each class's confidence score is computed using the softmax function [52], with the calculation principle defined as Equation (11):

$$Softmax(Z)_i = \frac{e^{z_i}}{\sum_{j=1}^{c} e^{z_j}} \tag{11}$$

where z represents the input vector, C denotes the number of classes in the dataset, and e^{z_i} and e^{z_j} are the standard exponential functions of the input and output vectors, respectively. The predicted class label obtained through the confidence averaging method is computed based on the following Equation (12):

$$\hat{y} = arg \max_{i\varepsilon 1,\cdots,c} \frac{\sum_{j=1}^{k} p_j(y = i \mid B_j, x)}{k} \tag{12}$$

where B_j represents the j -th base learner, and k denotes the number of selected base CNN classifiers. In the proposed IDS, $k = 5$; $p_j(y = i \mid B_j, x)$ indicates the prediction confidence of a class value i in a data sample x using B_j. Unlike traditional voting methods that consider only class labels, confidence averaging allows the ensemble model to rectify misclassified samples by leveraging classification confidence scores [53]. The integration strategy employed in this study is confidence averaging, which enables the ensemble model to identify uncertain classification results and rectify misclassified samples by leveraging classification confidence scores.

IV. EXPERIMENT AND RESULT ANALYSIS

This section provides an in-depth analysis and discussion of the proposed model across different datasets, comparing its performance with current mainstream models. To establish a comprehensive benchmark, we introduce Bayesian Optimization combined with the Tree-structured Parzen Estimator (BO-TPE) as a comparative method. Additionally, the experimental design includes a model variant without BKA to emphasize its impact on hyperparameter optimization. Furthermore, to thoroughly evaluate the effectiveness of the transfer learning approach, we incorporate a CNN model based on the LeNet-5 architecture trained from scratch and contrast its performance with models employing transfer learning techniques.

A. EXPERIMENTAL ENVIRONMENT AND PERFORMANCE INDICATORS

The experimental setup is configured as follows: the processor used is an Intel(R) Xeon(R) Gold 5218R CPU with a base clock speed of 2.10 GHz, running on the Windows 10 operating system. The system is equipped with 128 GB of RAM and an NVIDIA RTX 4090 GPU with 24 GB of dedicated video memory.

The performance evaluation of the model is conducted using four key metrics. Accuracy measures the proportion of correctly classified positive and negative samples relative to the total sample size, serving as the most intuitive indicator of model performance [34]. The calculation formula is provided in Equation (13). Precision quantifies the proportion of correctly predicted positive instances among all detected positives, as defined in Equation (14). Recall represents the ratio of correctly identified positive instances to the total actual positives, with its formulation given in Equation (15). Lastly, the F1-score, a weighted harmonic mean of Precision and Recall, is particularly advantageous in scenarios with imbalanced class distributions [34], and its computation is detailed in Equation (16).

$$Accuracy = \frac{TP + TN}{TP + FP + FN + TN} \tag{13}$$

$$Precision = \frac{TP}{TP + FP} \tag{14}$$

$$Recall = \frac{TP}{TP + FN} \tag{15}$$

$$F_1 = 2 \times \frac{precision \times recall}{precision + recall} \tag{16}$$

Furthermore, to evaluate the model from multiple perspectives, the testing time has been incorporated as one of the key performance metrics in the model assessment framework.

B. PERFORMANCE ANALYSIS OF CBCTL-IDS ON ToN-IoT DATASET

Figure 8 illustrates the training process of five models MobileNet, EfficientNet, Xception, VGG19, and Inception on the ToN-IoT dataset. The results indicate that both training accuracy and validation accuracy exceed 99%. Meanwhile, training and validation loss are reduced to below 0.1. The accuracy and loss curves for the WSN-DS and Edge-IIoT datasets exhibit similar trends; therefore, they are not redundantly presented in this paper.

TABLE 4. Comparison of the proposed model's performance against baseline models on the ToN-IoT dataset.

Model	Accuracy (%)	Precision (%)	Recall (%)	F1-score (%)	Test Time (s)
CNN	98.11	98.11	98.11	98.11	221
EfficientNet-Optimized	99.63	99.65	99.63	99.63	367
MobileNet-Optimized	99.63	99.63	99.63	99.63	259
VGG19-Optimized	99.43	99.47	99.42	99.44	297
Inception-Optimized	99.43	99.36	99.33	99.34	306
Xception-Optimized	99.53	99.43	99.43	99.43	331
CBCTL-IDS (without BKA)	98.79	98.97	98.97	98.97	356
CBCTL-IDS (with BO-TPE)	99.68	99.68	99.68	99.68	219
CBCTL-IDS	**99.85**	**99.85**	**99.85**	**99.85**	**209**

Table 4 presents the evaluation results of the proposed model on the ToN-IoT dataset in terms of accuracy, precision, recall, F1-score, and testing time. In terms of performance, the CBCTL-IDS model demonstrates the highest accuracy, reaching 99.85%, outperforming all baseline models. EfficientNet-Optimized and MobileNet-Optimized follow closely, both achieving an accuracy of 99.63%. The CNN model attains 98.11%, which, while respectable, falls short compared to the optimized models. The CBCTL-IDS (without BKA) variant exhibits a lower accuracy of 98.79%, highlighting the positive impact of the BKA optimization algorithm on model performance. Regarding testing time, CBCTL-IDS achieves the shortest inference time of 209 seconds, indicating its computational efficiency while maintaining high accuracy. The CNN model also performs well, with a testing time of 221 seconds. However, EfficientNet-Optimized exhibits the longest testing duration, requiring 367 seconds. Overall, the CBCTL-IDS model excels in multiple aspects, demonstrating superior accuracy while maintaining high computational efficiency. Experimenting on the ToN-IoT dataset, BO-TPE takes 7012 seconds to find hyperparameters while BKA takes 6496 seconds to find hyperparameters. The results show that BKA is more efficient. On the ToN-IoT dataset, a comparative analysis with mainstream deep learning models is presented in Table 5. The performance evaluation reveals that CBCTL-IDS achieves an accuracy of 99.85%, surpassing all competing models and establishing itself as the highest-performing approach among the compared methods. For reference, the accuracy of competing models is as follows: TBPTT [2] achieves 99.50%, XSRU-IoMT [54] attains 99.38%, SATIDS [55] records 96.56%, and FT-Transformer [56] yields 97.06%. Notably, CBCTL-IDS outperforms its closest competitor, TBPTT [2], by 0.35%, and exhibits a more than 3% improvement over SATIDS [55] and FT-Transformer [56]. In terms of test time, CBCTL-IDS requires 209 seconds, which, although longer than TBPTT [19] at 101 seconds, remains acceptable given its superior accuracy, precision, recall, and F1 score. The extended inference time is likely attributed to the increased computational complexity introduced by the ensemble-based architecture of the proposed model. Figure 9 illustrates the ten-class confusion matrix of the proposed model on the ToN-IoT dataset. The model exhibits a high degree of alignment between predicted and actual classifications. Notably, it demonstrates strong performance in identifying prevalent network threats in smart agriculture, such as DDoS, DoS, Ransomware, and Scanning attacks.

TABLE 5. Performance evaluation of mainstream models and the proposed model on the ToN-IoT dataset.

Model	Accuracy (%)	Precision (%)	Recall (%)	F1-score (%)	Test Time (s)
TBPTT [2]	99.50	99.31	99.24	99.39	101
SATIDS [55]	96.56	97.30	97.35	97.40	-
FT-Transformer [56]	97.06	97.23	96.67	96.94	-
XSRU-IoMT [54]	99.38	99.38	99.38	99.38	-
CBCTL-IDS	**99.85**	**99.85**	**99.85**	**99.85**	**209**

C. PERFORMANCE ANALYSIS OF CBCTL-IDS ON WSN-DS DATASET

Table 6 presents a comparative analysis of the proposed CBCTL-IDS model against the baseline CNN model in terms of performance and computational cost. The CBCTL-IDS model achieves an impressive 99.70% accuracy, precision, recall, and F1-score, significantly outperforming other baseline models, particularly in high-precision detection tasks. Although MobileNet-Optimized exhibits the shortest test time (256 seconds), its overall performance remains slightly inferior to CBCTL-IDS. Similarly, while CBCTL-IDS (with BO-TPE) achieves results close to those of CBCTL-IDS but falls marginally short across all evaluation metrics.

This comparison highlights the superior impact of BKA over BO-TPE in optimizing model performance. In terms of computational efficiency, CBCTL-IDS shows a moderate test time of 306 seconds. It outperforms CNN and EfficientNet-Optimized, reducing test time by 162 seconds and 215 seconds, respectively. However, it remains 50 seconds slower than MobileNet-Optimized. Experiments conducted on the WSN-DS dataset found that BO-TPE takes 7812 seconds to find hyperparameters, while BKA takes 7616 seconds to find hyperparameters. The results show that BKA is more efficient.

The proposed CBCTL-IDS model was trained and evaluated using the imbalanced WSN-DS dataset, with results

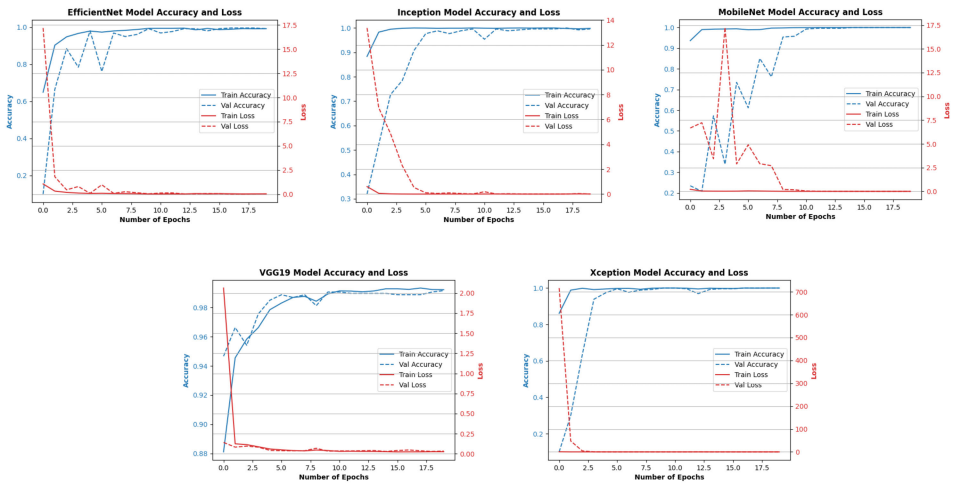

FIGURE 8. Training and validation accuracy and loss curves for the ToN-IoT dataset.

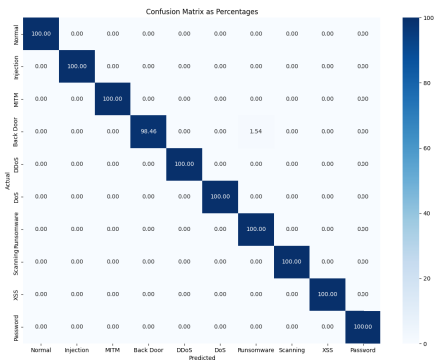

FIGURE 9. Confusion matrix of CBCTL-IDS on the ToN-IoT dataset.

TABLE 6. Comparison of the proposed model's performance against baseline models on the WSN-DS dataset.

Model	Accuracy (%)	Precision (%)	Recall (%)	F1 (%)	Test Time
CNN	98.01	98.01	98.01	98.01	468
EfficientNet-Optimized	99.07	99.05	99.07	99.03	521
MobileNet-Optimized	99.52	99.53	99.53	99.53	256
VGG19-Optimized	99.29	99.36	99.28	99.30	402
Inception-Optimized	99.40	99.40	99.40	99.40	369
Xception-Optimized	99.33	99.33	99.33	99.33	357
CBCTL-IDS (without BKA)	98.05	98.05	98.05	98.05	317
CBCTL-IDS (with BO-TPE)	99.58	99.56	99.56	99.56	302
CBCTL-IDS	**99.70**	**99.71**	**99.70**	**99.70**	**306**

presented in Table 7. The model achieved an impressive 99.70% accuracy, 99.71% precision, 99.70% recall, and 99.70% F1-score, outperforming current mainstream approaches. Specifically, compared to CNN-LSTM [57], our model demonstrates a 1.27% improvement in accuracy, underscoring its superior feature extraction capabilities in intrusion detection. Furthermore, it surpasses SCNN-Bi-LSTM [58] by 0.20% in accuracy, indicating enhanced generalization ability while maintaining a low false positive

TABLE 7. Performance evaluation of mainstream models and the proposed model on the WSN-DS dataset.

Model	Accuracy (%)	Precision (%)	Recall (%)	F1-score (%)	Test Time (s)
KNN-AOA [59]	95.65	90.83	82.77	80.78	-
CNN-LSTM [57]	98.43	99.12	98.03	98.32	-
SCNN-Bi-LSTM [58]	99.50	99.50	99.40	99.40	-
CBCTL-IDS	**99.70**	**99.71**	**99.70**	**99.70**	**306**

rate (FP) and false negative rate (FN). The improvements in accuracy (99.71%) and recall (99.70%) highlight the model's efficacy in significantly reducing both false alarms and missed detections. In terms of computational cost, the proposed model completes testing in 306 seconds.

Figure 10 presents the multi-class confusion matrix of the proposed model on the WSN-DS dataset. Despite the dataset's imbalanced nature, the model exhibits strong classification performance. Notably, it achieves an accuracy of 91.30% in detecting the Flooding attack, a prevalent security threat in smart agriculture environments.

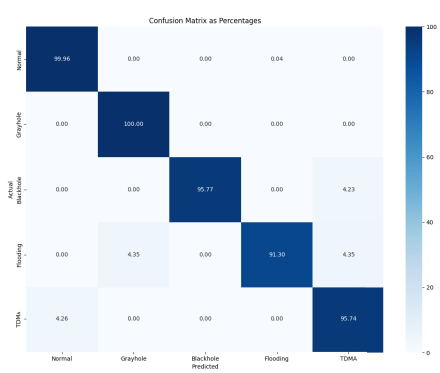

FIGURE 10. Confusion matrix of CBCTL-IDS on the WSN-DS dataset.

TABLE 8. Comparison of the proposed model's performance against baseline models on the Edge_IIoT dataset.

Model	Accuracy (%)	Precision (%)	Recall (%)	F1-score (%)	Test Time (s)
CNN	97.10	97.10	97.10	97.10	562
EfficientNet-Optimized	99.10	99.21	99.10	99.05	432
MobileNet-Optimized	99.58	99.59	99.58	99.58	422
VGG19-Optimized	98.60	98.92	98.60	98.63	596
Inception-Optimized	99.28	99.30	99.28	99.27	612
Xception-Optimized	99.29	99.29	99.29	99.29	562
CBCTL-IDS (without BKA)	97.32	97.32	97.32	97.32	510
CBCTL-IDS (with BO-TPE)	99.71	99.72	99.70	99.71	562
CBCTL-IDS	**99.90**	**99.88**	**99.90**	**99.89**	**462**

TABLE 9. Performance evaluation of mainstream models and the proposed model on the Edge_IIoT dataset.

Model	Accuracy (%)	Precision (%)	Recall (%)	F1-score (%)	Test Time (s)
TBPTT [2]	98.32	98.78	98.16	97.22	69
SACNN [60]	99.99	99.66	99.11	99.37	19
FFL-IDS [61]	97.00	93.00	96.00	92.60	-
ALNS [62]	94.98	95.30	93.01	97.71	-
CBCTL-IDS	**99.90**	**99.88**	**99.90**	**99.89**	**462**

D. PERFORMANCE ANALYSIS OF CBCTL-IDS ON EDGE_IIoT DATASET

To further validate the effectiveness of the proposed model, additional experiments were conducted on the Edge_IIoT dataset, which contains a larger volume of data. The comparative performance analysis between the proposed model and baseline models on Edge_IIoT is presented in Table 8.

From a performance perspective, CBCTL-IDS achieves an accuracy of 99.90%, the highest among all evaluated baseline models. The CBCTL-IDS (with BO-TPE) variant achieves a comparable accuracy of 99.71%, though slightly lower. Meanwhile, CBCTL-IDS (without BKA) exhibits a significantly reduced accuracy of 97.32%, demonstrating an evident performance gap compared to CBCTL-IDS and CBCTL-IDS (with BO-TPE). Notably, all other baseline models perform worse than CBCTL-IDS and CBCTL-IDS (with BO-TPE). In terms of optimization strategies, the BKA algorithm proves more effective than BO-TPE, enhancing the accuracy from 97.32% to 99.90%. Experiments on the Edge-IIoT dataset found that BO-TPE takes 25,432 seconds to find the hyperparameters while BKA takes 21,493 seconds to find the hyperparameters. The results show that BKA is more efficient. Regarding testing time, CBCTL-IDS achieves a moderate execution time of 462 seconds. The CBCTL-IDS (with BO-TPE) variant requires 562 seconds, slightly exceeding that of CBCTL-IDS, while CBCTL-IDS (without BKA) takes 510 seconds, which is also marginally longer. Other baseline models exhibit extended testing durations, particularly Inception-Optimized (612 seconds) and VGG19-Optimized (596 seconds). By integrating the efficient BKA optimization strategy, CBCTL-IDS demonstrates superior overall performance, establishing itself as the most effective model among the evaluated approaches.

The performance comparison between the proposed CBCTL-IDS model and mainstream models on the Edge-IIoT dataset is presented in Table 9. In terms of accuracy, CBCTL-IDS achieves 99.90%, slightly lower than SACNN [60]. However, it outperforms all counterparts in precision (99.88%), recall (99.90%), and F1-score (99.89%), demonstrating superior overall detection capability. Regarding computational efficiency, CBCTL-IDS exhibits a longer test time, which can be attributed to the integration of multiple CNN models and ensemble strategies. Nevertheless, this trade-off results in enhanced comprehensive detection performance.

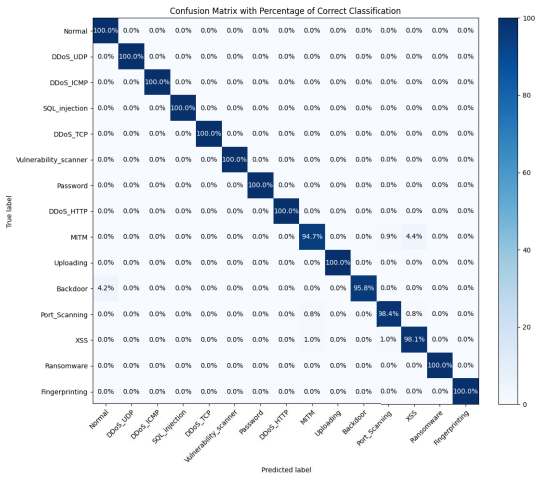

FIGURE 11. Presents the confusion matrix of CBCTL-IDS on the Edge-IIoT dataset.

Figure 11 presents the confusion matrix of CBCTL-IDS on the Edge-IIoT dataset. The system demonstrates outstanding predictive performance against common cybersecurity threats in smart agriculture, such as DDoS, SQL Injection, and Ransomware. However, the classification performance for MITM is relatively lower, which may be attributed to the limited number of samples available for this category.

V. CONCLUSION AND FUTURE DIRECTIONS

Over the past decade, SA IoT has leveraged advancements in IoT, artificial intelligence, and communication technologies to significantly enhance crop yields. However, the widespread deployment of interconnected devices within these systems renders them highly susceptible to malicious cyberattacks, which pose substantial security risks to agricultural production. Existing IoT-based IDS suffer from limitations, such as poor dataset generalization and insufficient detection accuracy, underscoring the urgent need to fortify the security of SA IoT systems.

To address these challenges, this study proposes CBCTL-IDS, an IDS based on deep transfer learning, and evaluates its

performance on three benchmark IoT security datasets: ToN-IoT, Edge-IIoTset, and WSN-DS. The proposed system first employs ET algorithm to extract features highly correlated with attack behaviors and transforms tabular data into image patches for input into five pre-trained CNN models via transfer learning. Subsequently, BKA is utilized to optimize hyperparameters, and the three most effective models are selected for confidence-based ensemble integration, yielding the final detection model. Experimental results demonstrate that the proposed framework achieves over 99% detection accuracy across both balanced and imbalanced datasets, effectively identifying anomalous traffic in IoT environments and improving the security of SA IoT systems.

In this study, the integration of multiple models has resulted in a relatively increased level of complexity. Several promising directions for future improvements are worth exploring: First, model lightweighting should be prioritized to adapt to resource-constrained agricultural IoT devices, aiming to reduce the computational and memory footprint of the models.Second, the exploration of federated learning could significantly enhance data privacy preservation and system performance in dynamic IoT environments. Third, the development of a hybrid IDS that combines rule-based and anomaly-based approaches is a noteworthy avenue for further research. Fourth, investigating diverse feature extraction algorithms and ensemble techniques could further improve the robustness and accuracy of the system.

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